Dataflow Design Module

Time Craft

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DFD Level 0 (context level diagram)

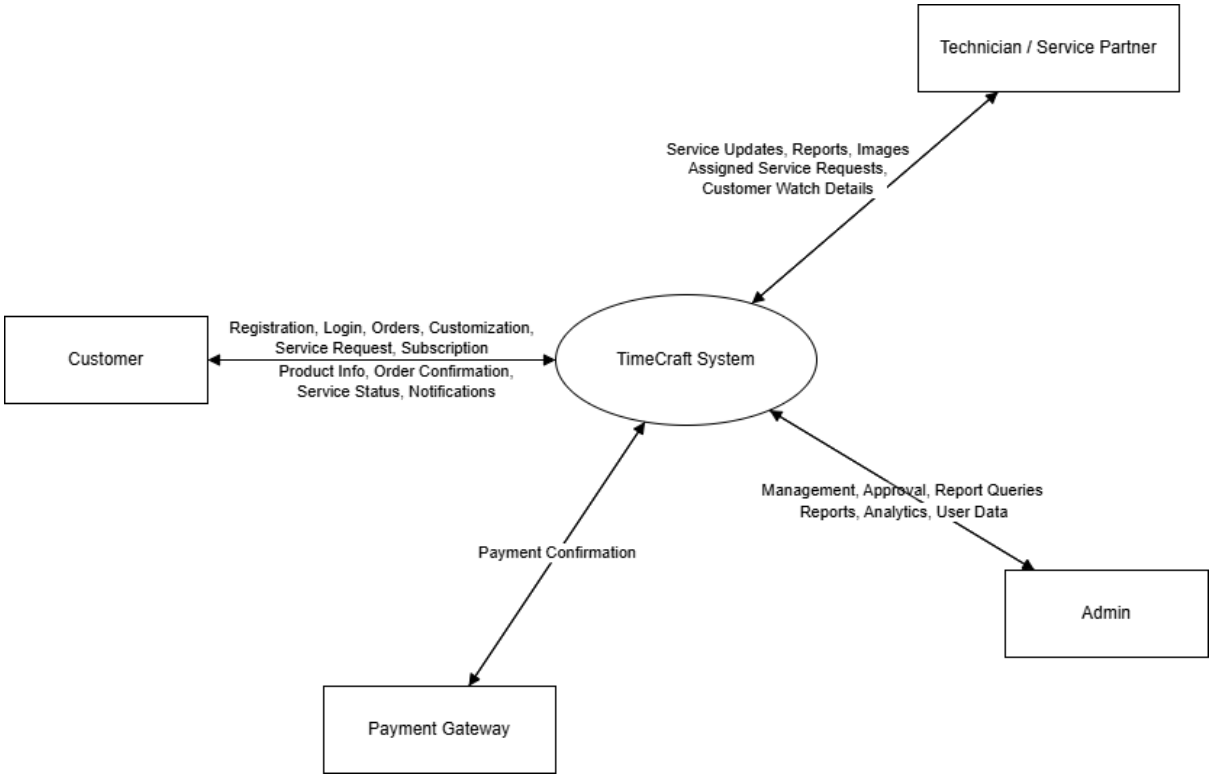
Entities:

1. Customer

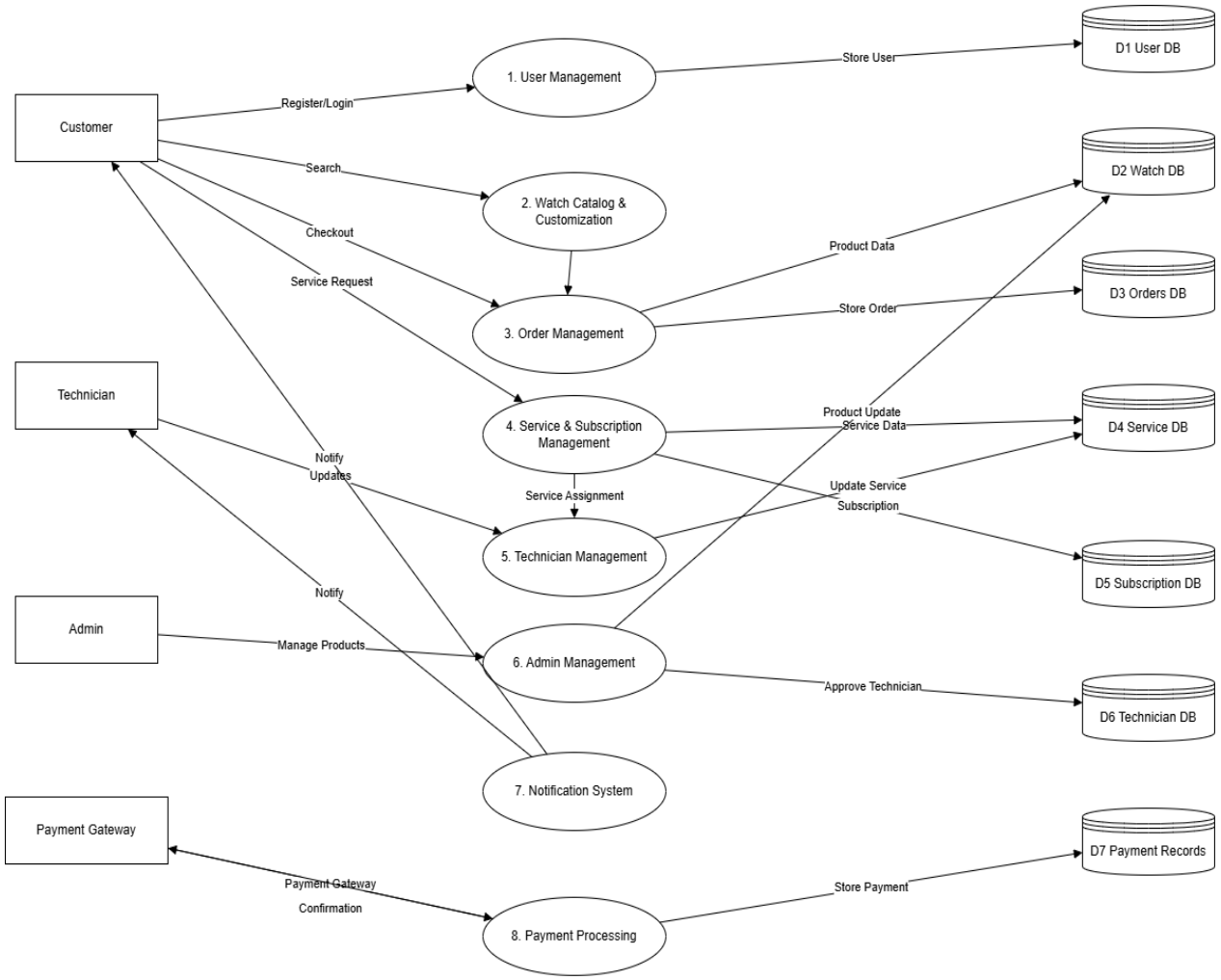
2. Technician / Service Partner

3. Admin

4. Payment Gateway

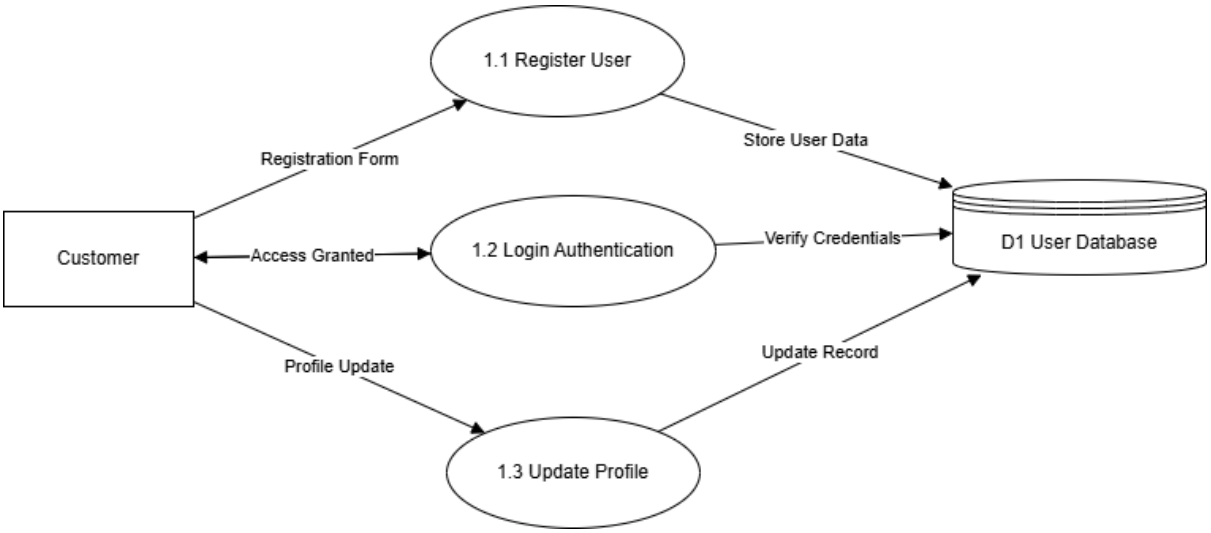


DFD Level 1

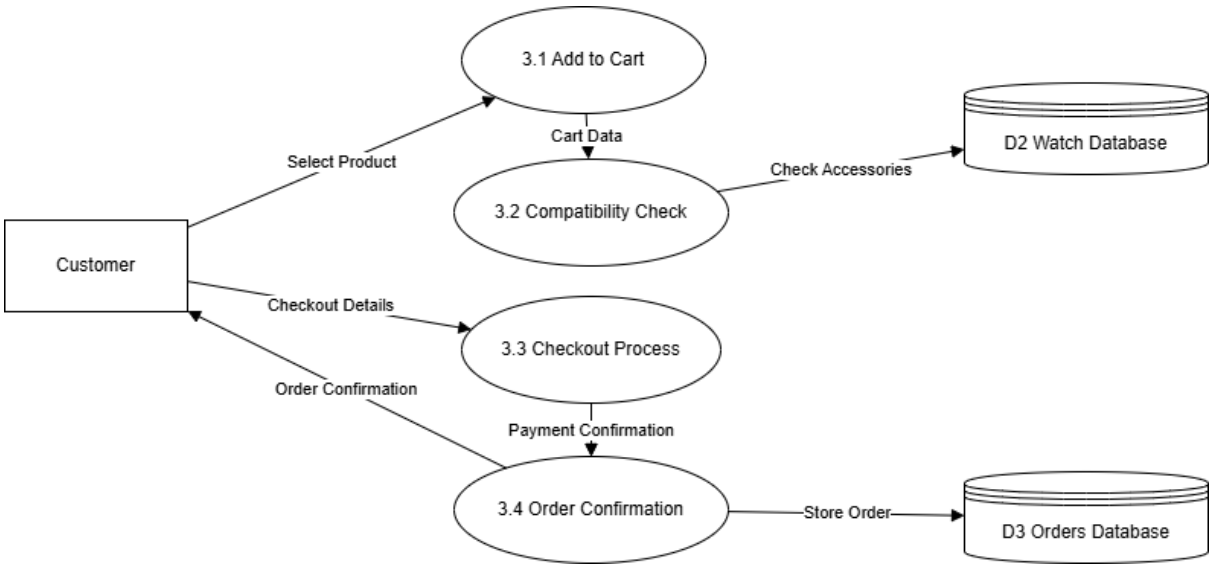

Customer
Technician
Admin
Payment Gateway
Register/Login
Search
Checkout
Service Request
Notify
Updates
Notify
Manage Products
Payment Gateway
Confirmation
1. User Management
2. Watch Catalog & Customization
3. Order Management
4. Service & Subscription Management
5. Technician Management
6. Admin Management
7. Notification System
8. Payment Processing
Service Assignment
Store User
Product Data
Store Order
Product Update
Service Data
Update Service
Subscription
Approve Technician
Store Payment
D1 User DB
D2 Watch DB
D3 Orders DB
D4 Service DB
D5 Subscription DB
D6 Technician DB
D7 Payment Records

DFD Level 2

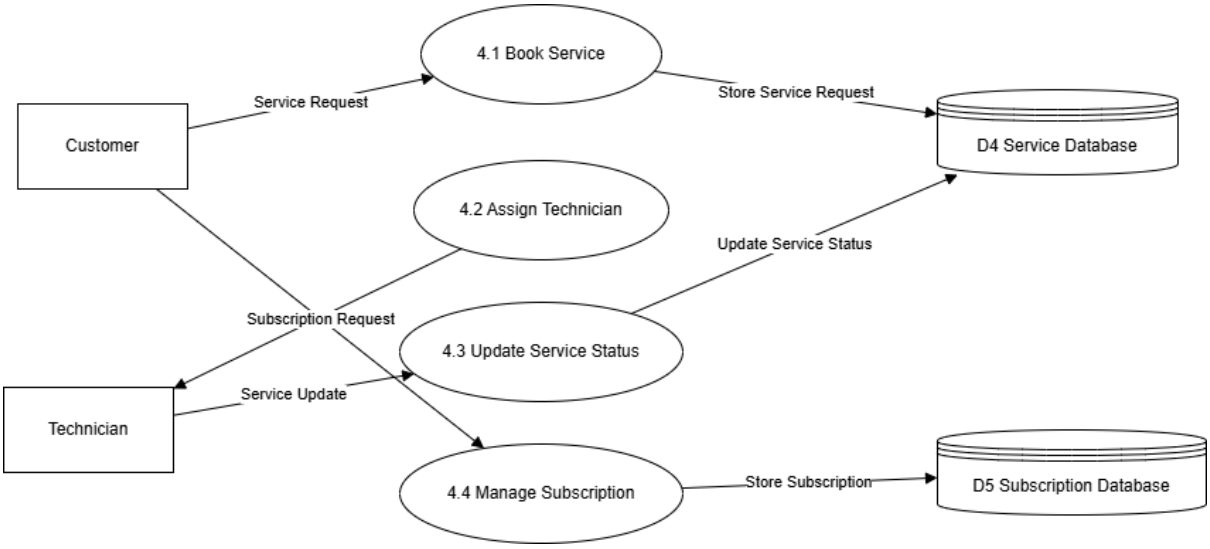
User Management



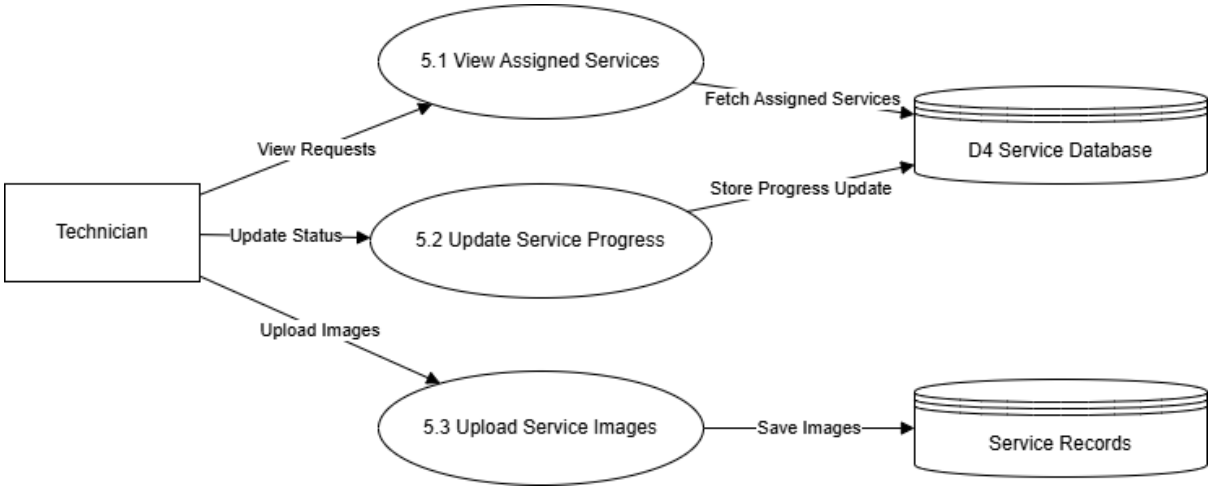
Order Management



Service & Subscription Management



Technician Management



Module Structure Chart

Requirements Traceability Matrix

Req ID	Requirement Description	Module	Sub-Module	Output
R1	User should be able to register	User Management	Register User	User account created
R2	User should login securely	User Management	Login Authentication	User dashboard access
R3	User can update profile	User Management	Update Profile	Updated user data
R4	User can browse watches	Watch Catalog	Browse Watches	Watch list displayed
R5	User can search watches	Watch Catalog	Search Watches	Filtered watch results
R6	User can view watch details	Watch Catalog	View Watch Details	Product information
R7	User can customize watch	Watch Catalog	Customize Watch	Customized watch preview
R8	User can add product to cart	Order Management	Add to Cart	Cart updated
R9	System checks accessory compatibility	Order Management	Check Compatibility	Compatible accessories
R10	User can checkout order	Order Management	Checkout Process	Order placed
R11	System confirms order	Order Management	Order Confirmation	Order ID generated
R12	User can book watch service	Service Management	Book Watch Service	Service request created
R13	System assigns technician	Service Management	Assign Technician	Technician assigned
R14	Technician updates service status	Service Management	Update Service Status	Service progress updated
R15	User can subscribe to plans	Service Management	Manage Subscription Plans	Subscription activated
R16	Technician views assigned jobs	Technician Management	View Assigned Services	Service list displayed
R17	Technician updates progress	Technician Management	Update Service Progress	Updated service status
R18	Technician uploads service images	Technician Management	Upload Service Images	Images stored

R19	System processes payment	Payment Processing	Process Payment	Payment confirmation
R20	System stores payment records	Payment Processing	Store Payment Records	Payment database updated
R21	System sends order notifications	Notification System	Order Notifications	Notification sent
R22	System sends service notifications	Notification System	Service Notifications	Notification sent
R23	System sends subscription notifications	Notification System	Subscription Notifications	Notification sent
R24	Admin manages products	Admin Management	Manage Products	Product updated
R25	Admin approves technicians	Admin Management	Approve Technicians	Technician activated
R26	Admin manages users	Admin Management	Manage Users	User management
R27	Admin generates reports	Admin Management	Generate Reports	System reports

Data Dictionary

User Table

Attribute	Data Type	Size	Constraints	Description
user_id	INT	11	Primary Key, Auto Increment	Unique user identifier
name	VARCHAR	50	NOT NULL	Full name of the user
email	VARCHAR	100	UNIQUE, NOT NULL	User email address
password	VARCHAR	60	NOT NULL	Encrypted user password
phone	VARCHAR	10	UNIQUE	User contact number
address	VARCHAR	150	NULL	User address
role	VARCHAR	20	DEFAULT 'Customer'	User role in system
created_at	DATETIME	—	NOT NULL	Account creation date
status	VARCHAR	10	DEFAULT 'Active'	Account status

Watch Table

Attribute	Data Type	Size	Constraints	Description
watch_id	INT	11	Primary Key, Auto Increment	Unique watch ID
watch_name	VARCHAR	60	NOT NULL	Name of the watch
brand	VARCHAR	40	NOT NULL	Brand name
price	DECIMAL	8,2	NOT NULL	Price of watch
description	VARCHAR	255	NULL	Watch description
category	VARCHAR	30	NOT NULL	Watch category
stock_quantity	INT	5	NOT NULL	Available quantity
image_url	VARCHAR	150	NULL	Watch image link
created_at	DATETIME	—	NOT NULL	Product creation date

Cart Table

Attribute	Data Type	Size	Constraints	Description
cart_id	INT	11	Primary Key, Auto Increment	Unique cart ID
user_id	INT	11	Foreign Key	User who owns cart
watch_id	INT	11	Foreign Key	Watch added to cart
quantity	INT	3	NOT NULL	Quantity of product
added_date	DATETIME	—	NOT NULL	Date item added

Order Table

Attribute	Data Type	Size	Constraints	Description
order_id	INT	11	Primary Key, Auto Increment	Unique order ID
user_id	INT	11	Foreign Key	Customer placing order
total_amount	DECIMAL	8,2	NOT NULL	Total order amount
order_status	VARCHAR	15	DEFAULT 'Pending'	Order status
order_date	DATETIME	—	NOT NULL	Order date
payment_status	VARCHAR	15	DEFAULT 'Pending'	Payment status

Service Request Table

Attribute	Data Type	Size	Constraints	Description
service_id	INT	11	Primary Key, Auto Increment	Unique service request ID
user_id	INT	11	Foreign Key	Customer requesting service
watch_id	INT	11	Foreign Key	Watch being serviced
service_type	VARCHAR	40	NOT NULL	Type of watch service
service_status	VARCHAR	15	DEFAULT 'Pending'	Current service status
request_date	DATETIME	—	NOT NULL	Date of service request

Technician Table

Attribute	Data Type	Size	Constraints	Description
technician_id	INT	11	Primary Key, Auto Increment	Unique technician ID
name	VARCHAR	50	NOT NULL	Technician name
email	VARCHAR	100	UNIQUE	Technician email
phone	VARCHAR	10	UNIQUE	Technician contact number
specialization	VARCHAR	50	NULL	Technician expertise
status	VARCHAR	10	DEFAULT 'Inactive'	Technician approval status

Payment Table

Attribute	Data Type	Size	Constraints	Description
payment_id	INT	11	Primary Key, Auto Increment	Unique payment ID
order_id	INT	11	Foreign Key	Related order ID
payment_method	VARCHAR	20	NOT NULL	Payment method used
amount	DECIMAL	8,2	NOT NULL	Payment amount
payment_status	VARCHAR	15	DEFAULT 'Pending'	Status of payment
payment_date	DATETIME	—	NOT NULL	Payment date

Subscription Table

Attribute	Data Type	Size	Constraints	Description
subscription_id	INT	11	Primary Key, Auto Increment	Unique subscription ID
user_id	INT	11	Foreign Key	Subscriber user ID
plan_name	VARCHAR	40	NOT NULL	Subscription plan
start_date	DATE	—	NOT NULL	Subscription start date
end_date	DATE	—	NOT NULL	Subscription expiry date
status	VARCHAR	10	DEFAULT 'Active'	Subscription status

Notification Table

Attribute	Data Type	Size	Constraints	Description
notification_id	INT	11	Primary Key, Auto Increment	Notification identifier
user_id	INT	11	Foreign Key	User receiving notification
type	VARCHAR	20	NOT NULL	Notification type
message	VARCHAR	255	NOT NULL	Notification message
sent_date	DATETIME	—	NOT NULL	Date notification sent